

M1 — Schema migration v4

Phase: 1 — Data **Status:** DONE (2026-05-26) **Blocked by:** M0 **Blocks:** M2, M14, M17, M19 **Estimated effort:** 1 day · **Actual:** 0.5 day

1. Purpose

Add the persistence layer for the V7 multi-patient extension: three new tables (`control_room`, `student`, `activity`) and five new columns on the existing `ehr_session` table. The migration is version-gated, idempotent under repeated runs, and preserves every v6 data row on upgrade — so an existing classroom-deployed v6 DB upgrades in place without operator intervention.

2. Structure

Files touched:

- `portal/ehr_db.py` — appends migration tuple (4, """"...""") to

`SCHEMA_MIGRATIONS`. Adds `CREATE TABLE` for `control_room`, `student`, `activity`; `ALTER TABLE` `ADD COLUMN` for the five new `ehr_session` columns; supporting indexes.

New tables:

Table	Purpose
<code>control_room</code>	One row per room. Holds <code>room_code</code> , <code>status</code> (active/frozen/ended), creation + end timestamps, optional Haiku rate / ElevenLabs char caps.
<code>student</code>	One row per learner registered to a room. <code>assigned_encounter_id</code> links to the encounter they're working in.
<code>activity</code>	Catalog of reusable case templates. <code>is_builtin=1</code> for the eight built-in activities, 0 for instructor-authored ones.

New `ehr_session` columns:

Column	Type	Default	Purpose
<code>room_id</code>	TEXT	NULL	Owning ControlRoom (NULL on legacy v6 rows).
<code>label</code>	TEXT	NULL	Instructor-visible tag, e.g. "Bed 3 — Kowalski".
<code>activity_id</code>	TEXT	NULL	Source Activity catalog entry, if any.
<code>chart_mode</code>	TEXT NOT NULL	'shared'	'shared' or 'private_clone'.
<code>patient_persona_id</code>	TEXT	NULL	Canonical persona library reference.

3. Uses

The migration runs automatically on first server start after the v7 upgrade — `ehr_db._open_db()` calls `_run_migrations()` which iterates `SCHEMA_MIGRATIONS` and applies any version greater than `MAX(schema_version)`. Subsequent starts no-op.

Downstream readers:

- M2 (`portal/control_room.py`) — the `ControlRoom/Student` dataclasses are the in-memory index over these tables.
- M8 — student-roster persistence reads/writes `student`.
- M11 — Activity catalog CRUD reads/writes `activity`.
- M14 — cohort debrief filters `ehr_session` by `room_id`.
- M17 — cost caps read `control_room.haiku_rate_cap` and `voice_char_cap`.
- M19 — capacity hardening counts rows in `ehr_session` by `room_id`.

4. Functions (exported API surface)

None. Migration 4 ships as a tuple inside `ehr_db.SCHEMA_MIGRATIONS` and is applied by the existing `ehr_db._run_migrations` runner. No new public functions in this module.

CRUD helpers for the new tables land in M8 (`student`) and M11 (`activity`), not here.

5. Limitations

- The `chart_mode` `ADD COLUMN` uses `DEFAULT 'shared'`. SQLite applies the default to existing rows lazily on read; new rows from M13 (private clone) explicitly write `'private_clone'`.
- No foreign-key constraints in the new tables. The codebase has historically used application-level integrity rather than SQLite FK enforcement; preserving that pattern.
- The migration does NOT backfill `room_id` for legacy v6 rows. Those rows remain `NULL` until the v7 wizard's "Open existing v6 session" flow lands (not in the current plan — a v7.1 candidate).
- `student.assigned_encounter_id` is a free `TEXT`, not constrained to a known `ehr_session.session_id`. M9 enforces validity at the route layer.

6. Test status

Test file	Asserts	Status	Last run
<code>tests/v7/test_migration_v4_idempotent.py</code>	Applying migrations twice is a no-op; <code>schema_version</code> row set unchanged on second pass.	PASS	2026-05-26
<code>tests/v7/test_migration_v4_preserves_v6_data.py</code>	A v6-snapshot DB (migrations 1-3 applied, sample <code>ehr_session</code> + <code>chart_event</code> rows) upgrades to v4 with legacy rows untouched, <code>NULL room_id</code> , default <code>chart_mode='shared'</code> , byte-identical <code>chart_event</code> payload.	PASS	2026-05-26

Test file	Asserts	Status	Last run
tests/v7/test_new_tables_exist.py	After full migration, control_room/student/activity tables exist with documented columns; ehr_session has all 5 new columns; supporting indexes present.	PASS	2026-05-26

7. Change list

Date	Author	Change	Files
2026-05-26	claude-code	Migration 4 appended with three new tables + five ehr_session columns + supporting indexes. 3 acceptance tests added under tests/v7/.	portal/ehr_db.py, tests/v7/test_migration_v4_idempotent.py, tests/v7/test_migration_v4_preserves_v6_data.py, tests/v7/test_new_tables_exist.py

8. Open questions / known issues

- `student.last_seen` is nullable; the v6 station-online heartbeat

pattern uses a non-null `last_seen`. M8 should pick a convention (likely: write `registered_at` to `last_seen` at create time and update on every student-bound request).

- `activity.seed_modules_json` is a JSON TEXT blob. Consider whether

a normalized `activity_module` join table would help M12 search, but until search performance is a problem, the JSON blob keeps the read path simple.

- The migration adds ONE `chart_mode NOT NULL DEFAULT 'shared'` column.

SQLite enforces NOT NULL strictly only on new INSERTs; existing rows take the default. Verify behavior on the LAN test in M21.